# (Supplementary Material) STAMPose: Spatial-Temporal Attention-Mamba for Efficient 3D Human Pose Estimation

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This supplementary material complements the main paper and is organized as follows:

1. per-action quantitative comparisons on Human3.6M (Sec. A);
2. zero-shot cross-dataset evaluation on 3DPW (Sec. B);
3. additional qualitative results on in-the-wild videos (Sec. C).

## A Per-Action Error Comparison

To assess the effectiveness of our method for each specific action, we present in Tab. 1 a per-action comparison of the MPJPE (P1) and P-MPJPE (P2) errors against existing methods on the Human3.6M dataset, using detected 2D pose sequences. Our findings indicate that STAMPose-L achieves the lowest P1 error on eight actions (Directions, Eating, Photo, Pose, Wait, WalkDog, Walk, and WalkTogether; tied on Wait), and attains the lowest average P1 among the methods in this table (37.9 mm). For the remaining actions, it stays close to the best results, whereas the sitting-type actions (Sitting, SittingDown) remain the most challenging. Overall, these results show that STAMPose maintains a competitive edge across a wide range of activities.

## B Zero-Shot Evaluation on 3DPW

To examine behavior beyond the training domain, we evaluate the Human3.6M-trained checkpoints of STAMPose-B, PoseMamba-B, and SasMamba-large in a zero-shot setting on the 3DPW test split (30 fps) [3, 4, 6]. 3DPW serves only as held-out test data; it is excluded from training, fine-tuning, and checkpoint selection.

**Protocol.** We use a custom scale-aligned protocol on a 14-joint subset obtained with a Human3.6M joint regressor; the resulting numbers are therefore best compared within Tab. 2 rather than with standard SMPL-based 3DPW results. Each checkpoint is evaluated with two 2D inputs: projected ground-truth 2D keypoints and the OpenPose-18 detections provided with 3DPW, mapped to the

Human3.6M 17-joint layout [1]. All models use $T$=243 with stride 81, symmetric edge padding, flip test-time augmentation, overlap averaging, and the same evaluator, and share the same valid-frame mask within each input condition. Missing detector keypoints are filled by within-track interpolation followed by a deterministic spatial fallback. We report person-frame-weighted N-MPJPE with per-frame scale alignment, PA-MPJPE with per-frame similarity alignment, and root-relative N-MPJVE with one scale per track, where N-MPJPE follows the scale-normalized formulation of Rhodin et al. [9].

**Results.** As shown in Tab. 2, STAMPose-B records the lowest N-MPJPE, PA-MPJPE, and N-MPJVE among the three models under both inputs. Its N-MPJPE is 22.48 mm lower than that of PoseMamba-B with OpenPose input and 12.62 mm lower with projected ground-truth input, while the PA-MPJPE gap to PoseMamba-B is small with projected ground-truth input (0.66 mm). We regard these single-checkpoint results as preliminary cross-dataset evidence; attributing the differences to specific factors such as camera motion, occlusion, the velocity loss $\mathcal{L}_m$, or DC-TSSM calls for additional controlled experiments.

## C Additional Qualitative Results on In-the-Wild Videos

To illustrate predictions beyond the studio conditions of the benchmark datasets, we apply STAMPose-L to in-the-wild videos unseen during training. Because ground-truth 3D annotations are unavailable for such videos, we visualize only the predicted 3D poses, with the left and right body sides drawn in red and blue, respectively.

Figure 1 shows the results across a range of everyday and athletic activities, including dancing, weightlifting, and basketball. Despite motion blur and large pose variations, STAMPose recovers structurally plausible 3D poses across these diverse in-the-wild scenes.

Figure 2 focuses on harder cases where the 2D detector visibly fails, using figure-skating, taekwondo, and snowboarding clips. Red arrows mark implausible 2D detections, such as abnormal limb lengths or swapped joints under fast motion, and missing joints under occlusion in the snowboarding clip; in many such cases, STAMPose still recovers a coherent 3D pose (green arrows). The rightmost column (taekwondo) shows a failure case: under an extreme high-kick pose, the 2D input is severely degraded, and this error propagates to the 3D estimate (red arrow on the 3D pose). This case relates to the robustness to imperfect 2D keypoints discussed in the limitations of the main paper.

**Table 1:** Per-action MPJPE (P1) and P-MPJPE (P2) comparison on Human3.6M using detected 2D pose sequences. $T$ is the number of input frames. In all tables, the best result is shown in **bold**, and the second-best result is underlined; tied results share the same mark.

| **MPJPE (P1)** | $T$ | Dire. | Disc. | Eat | Greet | Phone | Photo | Pose | Purch. | Sit | SitD | Smoke | Wait | WalkD | Walk | WalkT | **Avg** |
|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|
| MixSTE [12] | 243 | 37.6 | 40.9 | 37.3 | 39.7 | 42.3 | 49.9 | 40.1 | 39.8 | 51.7 | 55.0 | 42.1 | 39.8 | 41.0 | 27.9 | 27.9 | 40.9 |
| P-STMO [10] | 243 | 38.9 | 42.7 | 40.4 | 41.1 | 45.6 | 49.7 | 40.9 | 39.9 | 55.5 | 59.4 | 44.9 | 42.2 | 42.7 | 29.4 | 29.4 | 42.8 |
| StridedFormer [5] | 351 | 40.3 | 43.3 | 40.2 | 42.3 | 45.6 | 52.3 | 41.8 | 40.5 | 55.9 | 60.6 | 44.2 | 43.0 | 44.2 | 30.0 | 30.2 | 43.7 |
| STCFormer [11] | 243 | 39.6 | 41.6 | 37.4 | 38.8 | 43.1 | 51.1 | 39.1 | 39.7 | 51.4 | 57.4 | 41.8 | 38.5 | 40.7 | 27.1 | 28.6 | 41.0 |
| MotionBERT [13] | 243 | 36.6 | 39.3 | 37.8 | 33.5 | 41.4 | 49.9 | 37.0 | 35.5 | 50.4 | 56.5 | 41.4 | 38.2 | 37.3 | 26.2 | 26.9 | 39.2 |
| HDFormer [2] | 96 | 38.1 | 43.1 | 39.3 | 39.4 | 44.3 | 49.1 | 41.3 | 40.8 | 53.1 | 62.1 | 43.3 | 41.8 | 43.1 | 31.0 | 29.7 | 42.6 |
| MotionAGFormer-L [7] | 243 | 36.8 | 38.5 | 35.9 | 33.0 | 41.1 | 48.6 | 38.0 | 34.8 | 49.0 | **51.4** | 40.3 | 37.4 | 36.3 | 27.2 | 27.2 | 38.4 |
| KTPFormer [8] | 243 | 37.3 | 39.2 | 35.9 | 37.6 | 42.5 | 48.2 | 38.6 | 39.0 | 51.4 | 55.9 | 41.6 | 39.0 | 40.0 | 27.0 | 27.4 | 40.1 |
| PoseMamba-S [4] | 243 | 39.5 | 41.6 | 39.9 | 35.7 | 43.5 | 52.7 | 40.2 | 37.9 | 53.1 | 63.4 | 42.4 | 40.0 | 39.6 | 29.3 | 29.2 | 41.8 |
| PoseMamba-B [4] | 243 | 38.8 | 40.8 | 38.8 | 35.2 | 42.1 | 50.8 | 38.8 | 36.4 | 51.8 | 61.9 | 42.0 | 38.4 | 38.7 | 28.1 | 28.7 | 40.8 |
| PoseMamba-L [4] | 243 | 36.5 | **37.9** | 37.5 | 33.5 | **39.6** | 47.7 | 36.5 | **34.0** | **47.8** | 54.5 | **40.2** | **36.3** | 36.7 | 26.4 | 26.9 | 38.1 |
| SasMamba [3] | 243 | 39.0 | 43.0 | 39.2 | 36.1 | 43.2 | 51.3 | 40.0 | 37.2 | 52.7 | 61.7 | 43.1 | 38.9 | 39.8 | 28.3 | 28.6 | 41.5 |
| SasMamba-large [3] | 243 | 38.4 | 41.0 | 38.0 | 34.6 | 41.3 | 49.3 | 38.8 | 36.1 | 50.7 | 56.2 | 41.6 | 38.4 | 37.2 | 27.2 | 27.6 | 39.8 |
| STAMPose-S | 243 | 39.6 | 41.7 | 38.0 | 35.1 | 43.0 | 49.7 | 38.6 | 37.5 | 51.2 | 60.0 | 43.0 | 40.3 | 38.2 | 27.4 | 28.8 | 40.8 |
| STAMPose-B | 243 | 37.4 | 39.5 | 38.2 | **32.9** | 41.5 | 47.1 | 36.5 | 35.5 | 49.5 | 53.5 | 41.7 | 37.6 | 36.2 | 26.0 | 26.5 | 38.7 |
| STAMPose-L | 243 | **36.0** | 39.1 | **35.7** | 33.0 | 40.1 | **45.3** | **36.0** | 34.5 | 49.8 | 55.4 | 40.3 | **36.3** | **35.3** | **25.5** | **26.1** | **37.9** |

| **P-MPJPE (P2)** | $T$ | Dire. | Disc. | Eat | Greet | Phone | Photo | Pose | Purch. | Sit | SitD | Smoke | Wait | WalkD | Walk | WalkT | **Avg** |
|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|
| MixSTE [12] | 243 | 30.8 | 33.1 | 30.3 | 31.8 | 33.1 | 39.1 | 31.1 | 30.5 | 42.5 | **44.5** | 34.0 | 30.8 | 32.7 | 22.1 | 22.9 | 32.6 |
| P-STMO [10] | 243 | 31.3 | 35.2 | 32.9 | 33.9 | 35.4 | 39.3 | 32.5 | 31.5 | 44.6 | 48.2 | 36.3 | 32.9 | 34.4 | 23.8 | 23.9 | 34.4 |
| StridedFormer [5] | 351 | 32.7 | 35.5 | 32.5 | 35.4 | 35.9 | 41.6 | 33.0 | 31.9 | 45.1 | 50.1 | 36.3 | 33.5 | 35.1 | 23.9 | 25.0 | 35.2 |
| STCFormer [11] | 243 | **29.5** | 33.2 | 30.6 | 31.0 | 33.0 | 38.0 | 30.4 | 29.4 | 41.8 | 45.2 | **33.6** | **29.5** | 31.6 | **21.3** | **22.6** | 32.0 |
| MotionBERT [13] | 243 | 30.8 | 32.8 | 32.4 | 28.7 | 34.3 | 38.9 | 30.1 | 30.0 | 42.5 | 49.7 | 36.0 | 30.8 | 31.7 | 22.0 | 23.0 | 32.9 |
| HDFormer [2] | 96 | 29.6 | 33.8 | 31.7 | 31.3 | 33.7 | 37.7 | 30.6 | 31.0 | 41.4 | 47.6 | 35.0 | 30.9 | 33.7 | 25.3 | 23.6 | 33.1 |
| MotionAGFormer-L [7] | 243 | 31.0 | 32.6 | 31.0 | **27.9** | 34.0 | 38.7 | 31.5 | 30.0 | 41.4 | 45.4 | 34.8 | 30.8 | 31.3 | 22.8 | 23.2 | 32.5 |
| KTPFormer [8] | 243 | 30.1 | 32.3 | **29.6** | 30.8 | **32.3** | 37.3 | 30.0 | 30.2 | **41.0** | 45.3 | **33.6** | 29.9 | 31.4 | 21.5 | **22.6** | **31.9** |
| PoseMamba-S [4] | 243 | 32.1 | 34.8 | 34.1 | 30.7 | 36.0 | 41.7 | 33.0 | 32.1 | 44.5 | 53.8 | 36.5 | 32.3 | 34.2 | 25.0 | 25.2 | 35.0 |
| PoseMamba-B [4] | 243 | 32.3 | 34.0 | 33.2 | 30.2 | 34.9 | 40.6 | 32.0 | 31.0 | 44.5 | 53.0 | 36.3 | 31.3 | 33.5 | 23.8 | 24.6 | 34.3 |
| PoseMamba-L [4] | 243 | 31.0 | **32.2** | 32.5 | 28.5 | 33.1 | 38.6 | 30.5 | **29.2** | 41.3 | 47.8 | 34.9 | 30.2 | 31.7 | 22.7 | 23.5 | 32.5 |
| SasMamba [3] | 243 | 32.3 | 35.7 | 33.5 | 30.8 | 36.4 | 40.2 | 32.8 | 32.1 | 44.0 | 52.7 | 37.0 | 31.7 | 34.3 | 24.4 | 24.8 | 34.8 |
| SasMamba-large [3] | 243 | 31.9 | 34.0 | 32.4 | 29.8 | 34.7 | 39.2 | 31.7 | 31.3 | 43.2 | 49.5 | 36.2 | 31.1 | 32.2 | 23.5 | 23.9 | 33.6 |
| STAMPose-S | 243 | 33.0 | 34.5 | 32.8 | 30.0 | 35.8 | 38.9 | 31.7 | 32.2 | 43.4 | 52.8 | 37.1 | 32.2 | 32.7 | 23.2 | 24.5 | 34.3 |
| STAMPose-B | 243 | 31.7 | 33.4 | 32.8 | 28.4 | 35.0 | 38.0 | 30.1 | 31.3 | 42.9 | 47.5 | 36.5 | 30.7 | 31.1 | 22.3 | 23.2 | 33.0 |
| STAMPose-L | 243 | 30.8 | 32.9 | 31.0 | 28.4 | 33.0 | **37.2** | **29.9** | 30.1 | 42.6 | 47.5 | 35.0 | 29.6 | **30.3** | 21.7 | 22.7 | 32.2 |

**Table 2:** Zero-shot evaluation on the 3DPW test split with Human3.6M-trained checkpoints under our scale-aligned 14-joint protocol. N-MPJPE and PA-MPJPE are in mm, and N-MPJVE is in mm/s.

| **Model** | **2D input** | **N-MPJPE ↓** | **PA-MPJPE ↓** | **N-MPJVE ↓** |
|---|---|---|---|---|
| SasMamba-large | OpenPose | 202.68 | 118.97 | 251.14 |
| PoseMamba-B | OpenPose | 168.80 | 109.49 | 246.64 |
| STAMPose-B | OpenPose | **146.32** | **103.63** | **240.00** |
| SasMamba-large | Proj. GT | 131.41 | 74.16 | 204.79 |
| PoseMamba-B | Proj. GT | 110.48 | 65.17 | 200.15 |
| STAMPose-B | Proj. GT | **97.86** | **64.51** | **190.06** |

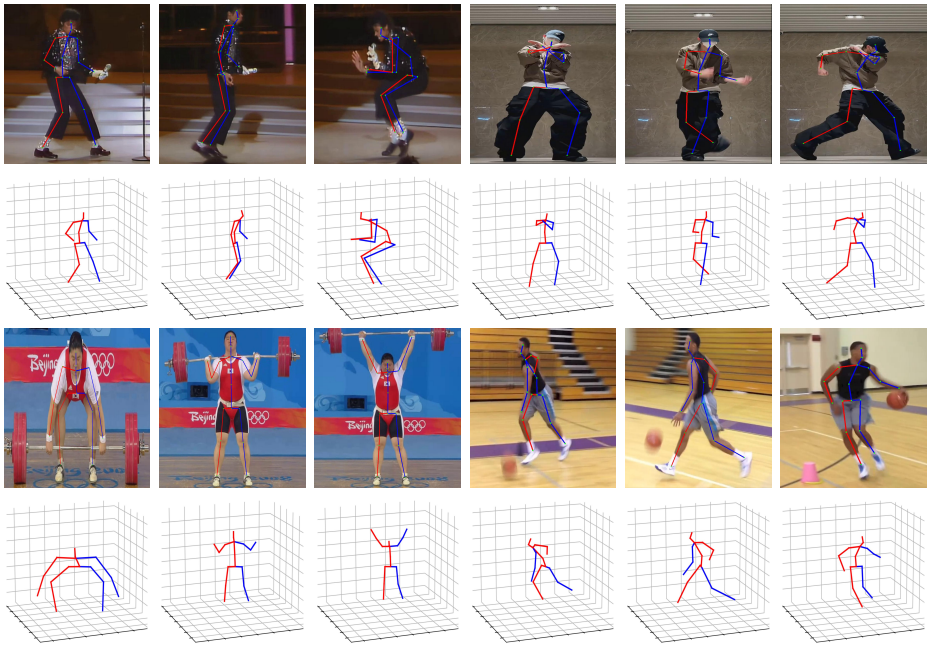


**Fig. 1:** Qualitative results of STAMPose on in-the-wild videos.

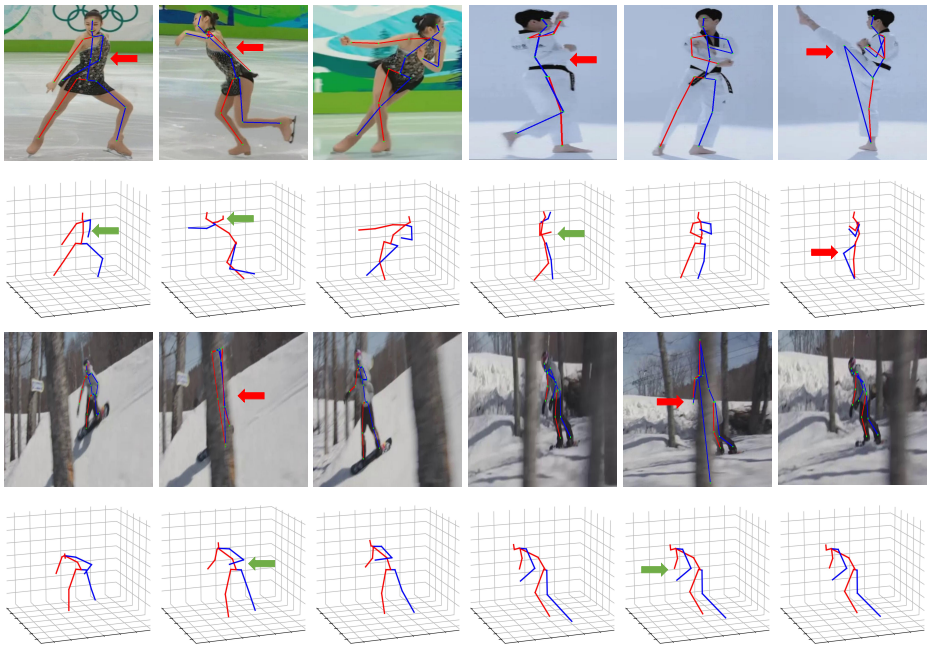


**Fig. 2:** Qualitative results of STAMPose on in-the-wild videos where the 2D detector fails.

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